

FERMENTATION PROCESSES

Sterilization, Recovery & Purification
of fermentation products

PART 1. STERILIZATION

PART 2. RECOVERY OF FERMENTATION PRODUCTS

PART 3. PURIFICATION OF FERMENTATION PRODUCTS

PART 1. STERILIZATION

INTRODUCTION

If a foreign microorganism invades the fermentation then the following consequences may occur:

1. Loss of productivity.
 2. If the fermentation is a continuous one then the contaminant may “outgrow” the production organism and displace it from the fermentation.
 3. Contaminate the final product
 4. The contaminant may produce compounds that make subsequent extraction of the final product difficult.
 5. The contaminant may degrade the desired product
 6. Contamination of a bacterial fermentation with phage could result in the lysis of the culture.
- Effect on economical benefits, health and environment.

INTRODUCTION

Avoidance of contamination may be achieved by

1. Effective design and construction of the fermentation plant
2. Using a pure inoculum to start the fermentation
3. Sterilizing the medium to be employed
4. Sterilizing the fermenter vessel
5. Sterilizing all materials to be added to the fermentation
6. Maintaining aseptic conditions during the fermentation
7. Putting in place detailed operating procedures for sterilization, aseptic maintenance, and staff training.

INTRODUCTION

Definition: *Sterilization* is a process to eliminate biological contamination in fermentation process:

- bacteria
 - yeast
 - fungi
 - bacteriophage
- **Other requirements to select sterilization methods/agents**
 - No much affect on quantity and quality of medium components
 - No production of inhibitors to fermentation microorganisms
 - No affect on human health, environment
 - Simply, easy in scaling up, reasonable price

INTRODUCTION

Two types of reaction contribute to the loss of nutrient quality during sterilization

Interactions between nutrient components of the medium

- A common occurrence during sterilization is the Maillard-type browning reaction that results in discoloration of the medium as well as loss of nutrient quality and the accumulation of growth-inhibitory compounds

Table 5.1 The Effect of Sterilization Time on Glucose Concentration and Product Accretion Rate in an Antibiotic Fermentation ([Corbett, 1985](#))

Time at 121°C (min)	Amount of Added Glucose Remaining (%)
60	35
40	46
30	64

INTRODUCTION

Two types of reaction contribute to the loss of nutrient quality during sterilization

Degradation of heat labile components

Certain vitamins, amino acids, and proteins may be degraded during a steam sterilization regime.

Table 5.1 The Effect of Sterilization Time on Glucose Concentration and Product Accretion Rate in an Antibiotic Fermentation ([Corbett, 1985](#))

Time at 121°C (min)	Amount of Added Glucose Remaining (%)
60	35
40	46
30	64

Rules to prevent contamination in fermentation

“3” NO

1. NO remaining: Clean, sterilize, filter... to ensure that there are no contaminated microbial cells left in the equipment or medium before culturing.
2. NO entering: Ensure that during operation there is no external contamination of the culture medium or equipment.
3. NO spreading: Prevent contamination from spreading in the work environment, or prevent spread once contamination has occurred.

Sources of contamination

Materials (including water)

High conc. of contaminated microorganisms

- yeast, fungi, bacteria, virus
- including also heat-resistant microorganisms
- including also spores

Sources of contamination

Equipment

- Inappropriate design
- Leakage in equipment
- Sensors work improperly

Sources of contamination

Operation

- From input flows (medium, microorganisms, aeration,...), sampling,...
- From the previous fermentation

Types of sterilization

- **Sterilization of media**
 - Components sterilized together
 - Components sterilized separately
- **Sterilization of equipment**
- **Sterilization of products**

STERILIZATION METHODS

- Physical methods
- Filtration methods
- Chemical methods

PHYSICAL METHODS

- Heat (moist heat or dry heat)
- Radiation

PHYSICAL METHODS - Heat

- Kinetics of viable cells v.s. sterilization time

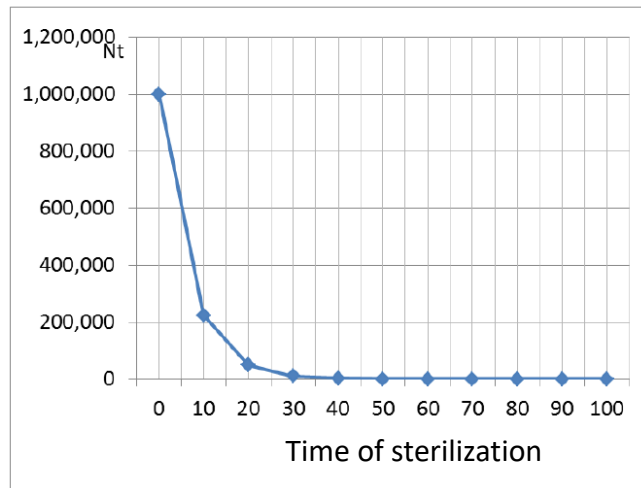
$$-dN/dt = k.N$$

$$\begin{aligned}\Rightarrow & -dN/N = k.dt \\ \Rightarrow & \int (-dN/N) = \int k.dt \\ \Rightarrow & \ln(N_t/N_o) = -kt \\ \Rightarrow & N_t/N_o = e^{-kt}\end{aligned}$$

N: the number of viable organisms present (N_o and N_t)

t: the time of sterilization (minutes)

k: the specific death rate (min^{-1}) (depend not only species, but also the physiological form of the cell)



$N_o = 1,000,000$ cells

$k = 0.15/\text{min}$

Calculate N_t when $t = 0; 50; 100; 200; \dots$

PHYSICAL METHODS - Heat

- **Advantages:**

- + popular

- + suitable for sterilization of media, equipment, products.

- **Disadvantages:**

- + high investment and operating costs

- + not suitable in some cases....

Note: Pasteurization vs. Sterilization

PHYSICAL METHODS – Heat

Dry heat

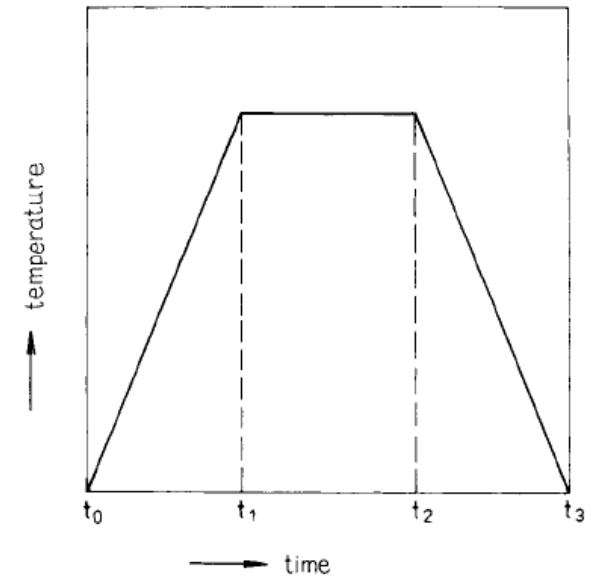
- Use drying oven
- 150 °C, 2h30; 160 °C, 2h; 170 °C, 1h
- **Advantages**
 - fast
 - easy handle
- **Disadvantages**
 - consume high energy
 - limited for liquid sterilization



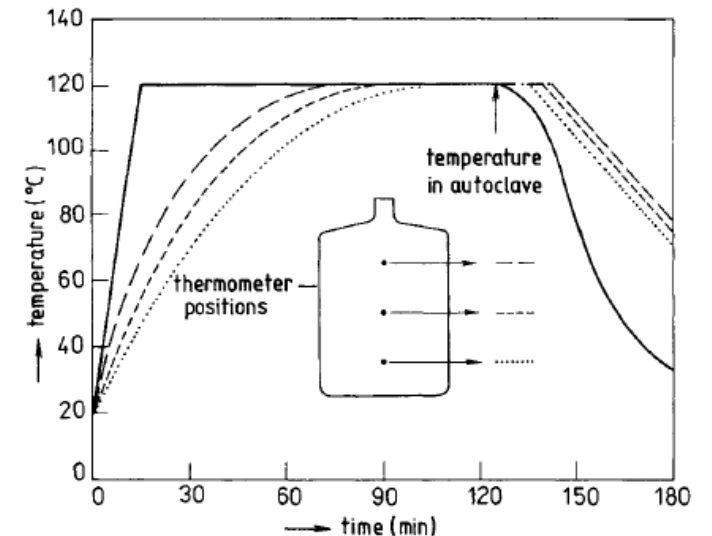
PHYSICAL METHODS – Heat

Moist heat

- Steam at high temp. (high pressure) kills microorganisms
- 121 °C, 15-20 min
- **Advantages**
 - suitable to various types of media/consumables
- **Disadvantages**
 - High investment and maintaining costs



Small volume of sterilization



Large volume of sterilization

PHYSICAL METHODS – Heat

Moist heat – lab scale

- 121 °C, 15-20 mins



- Some notes

PHYSICAL METHODS – Heat

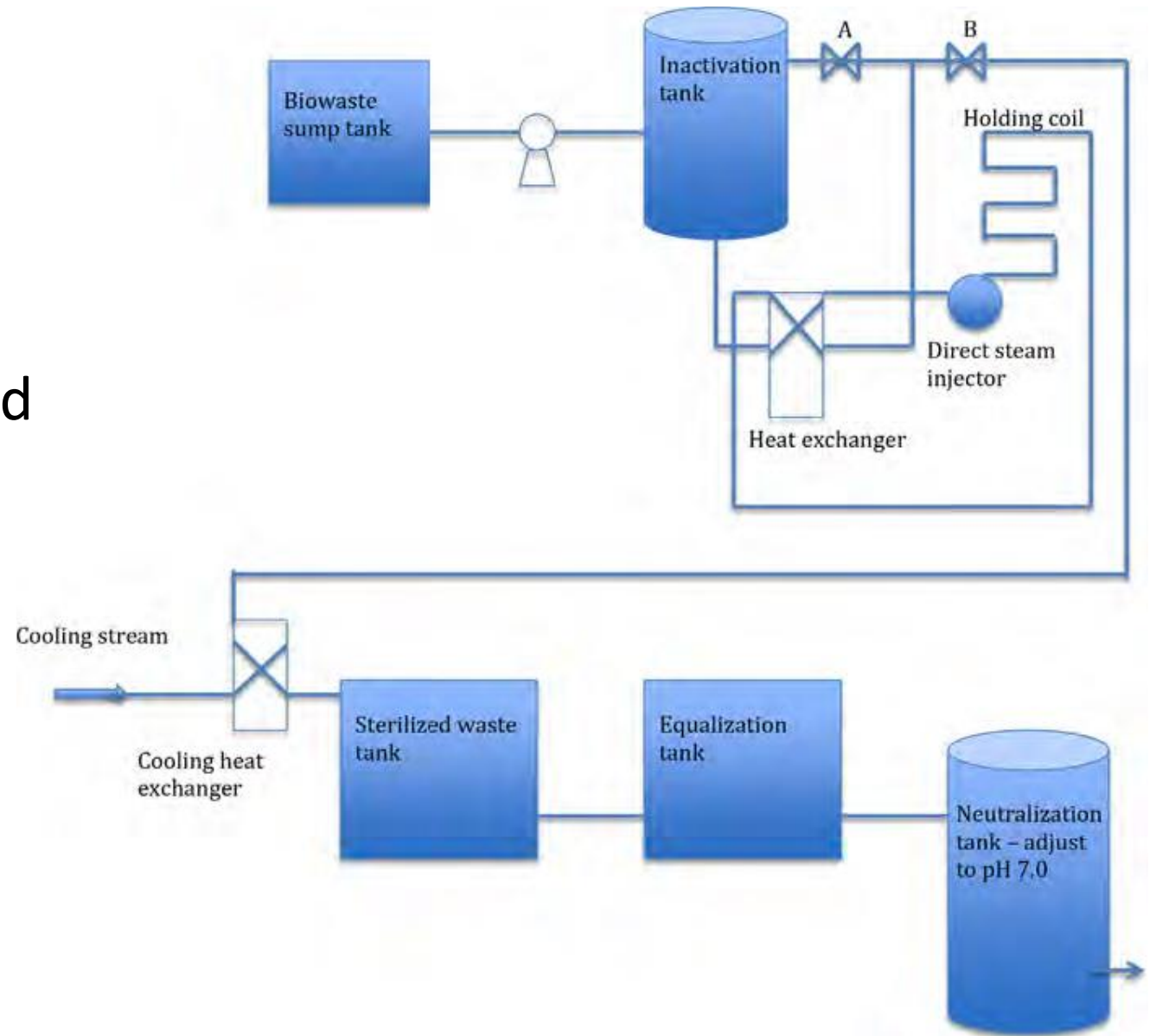
Moist heat – large scale

- The medium was sterilized in fermenter (121-125 °C)
- This is normally achieved by heating the jacket or coils of the fermenter with steam
- Then cooled down using the cooling system

Note: agitation is important...

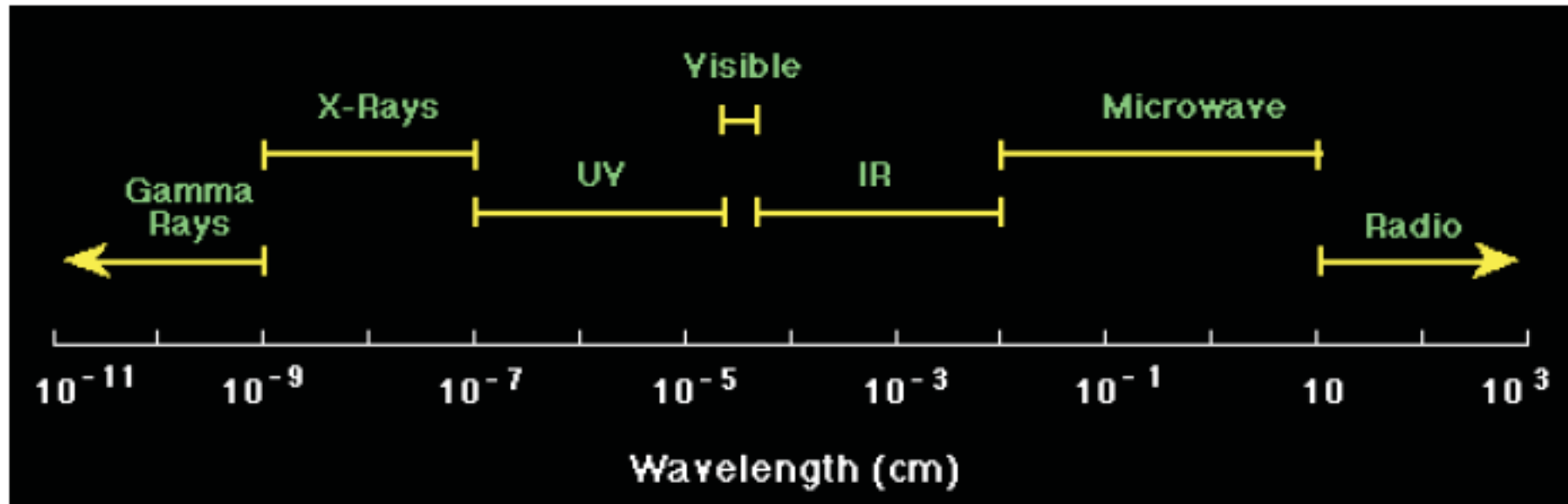
Sterilization of liquid wastes

- Waste biomass must be sterilized before disposal:
- Lab scale
- Larger scale



PHYSICAL METHODS - Radiation

- UltraViolet (UV)
- X rays and gamma rays



FILTRATION METHOD

- Both liquid and gas filtrations are utilized
- The removal of suspended solids from both gas and liquids in the filtration may be described by the following mechanisms:
 1. Inertial impaction (Va chạm quán tính)

The fluid in which the particles are suspended will flow through the filter by the route of least resistance.
 2. Diffusion (Sự khuếch tán)

Extremely small particles suspended in a fluid randomly move due to collisions with fluid molecules.

FILTRATION METHOD

- The removal of suspended solids from both gas and liquids in these processes may then be described by the following mechanisms:

3. Electrostatic attraction:

Charged particles may be attracted by opposite charges on the surface of the filtration medium

4. Interception (sự cản lại)

- Particles that are larger than the filter pores are removed by direct interception.
- However, a significant number of particles that are smaller than the filter pores are also retained by interception.

FILTRATION METHOD

- **Advantages**

- + fast, continuous
- + suitable in some cases, especially in air sterilization

- **Disadvantages**

- + high cost of filter membrane
- + small scale (*in liquid filtration*)
- + impossible to sterilize equipment

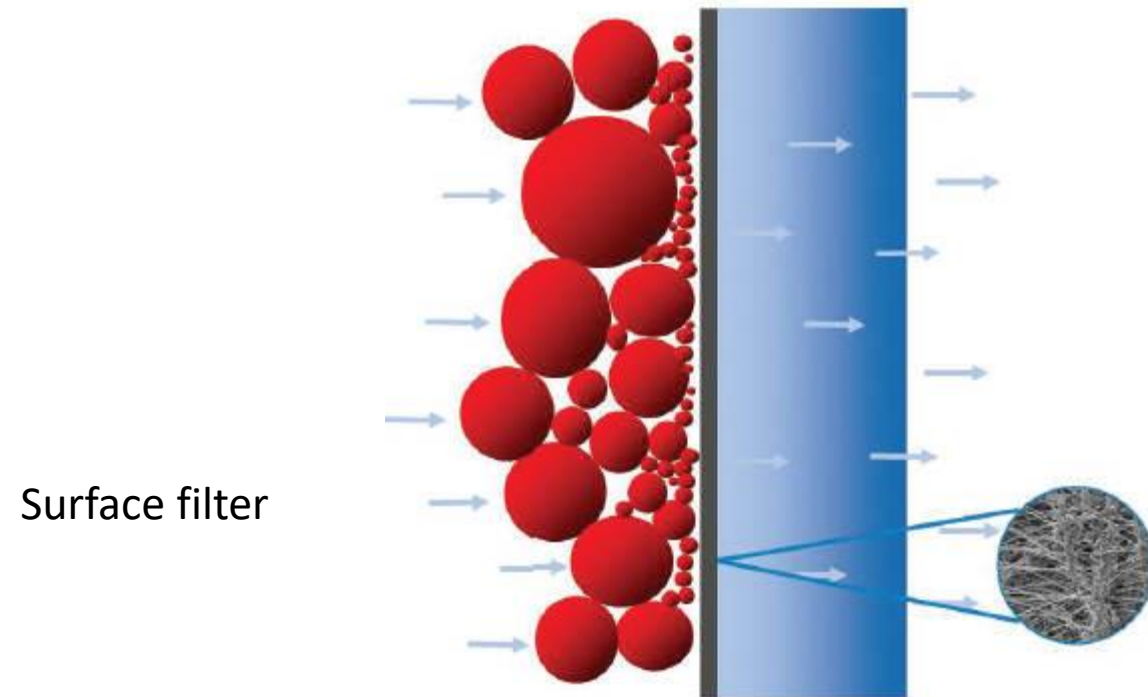
FILTRATION METHOD

Filters have been classified into two types

1. The pores in the filter are smaller than the particles

The particles are deposited on the outer surface of the medium and form a "cake".

Ex:



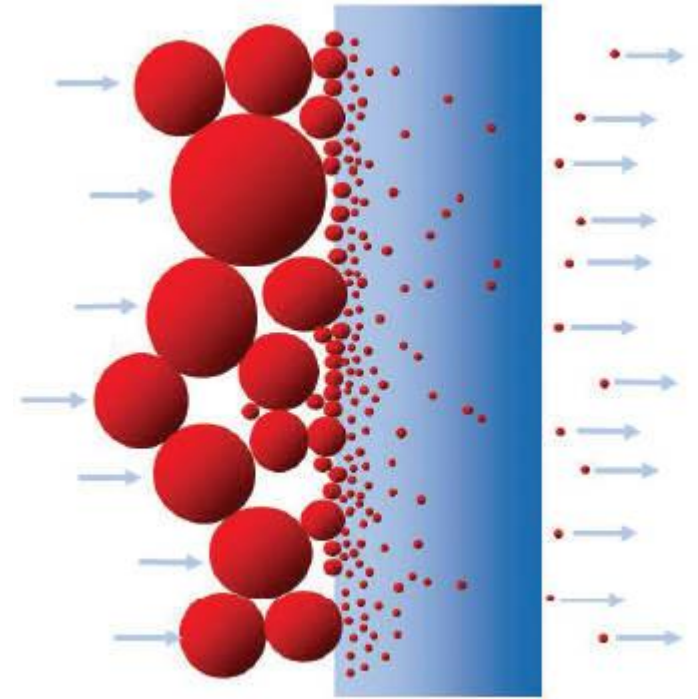
FILTRATION METHOD

Filters have been classified into two types

2. The pores in the filter are larger than the particles

Ex:

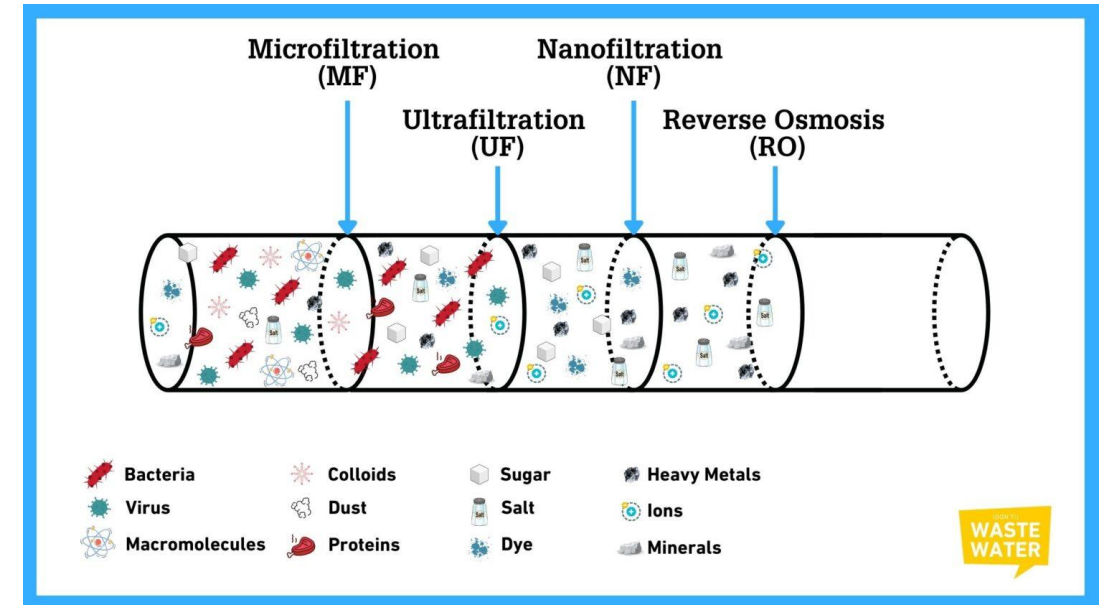
The filters, themselves, must be sterile...



Depth and non-fixed pore filter.

FILTRATION METHOD

Membrane	Approximate diameter range of molecules, ions or particles retained (µm)
Microfiltration	0.10 a 1.0
Ultrafiltration	0.01 a 0.10
Nanofiltration	0.001 a 0.01
Reverse osmosis (also called "hyperfiltration")	0.0001 a 0.001



Filter sterilization of liquids

The following liquids are sterilized in a fermentation:

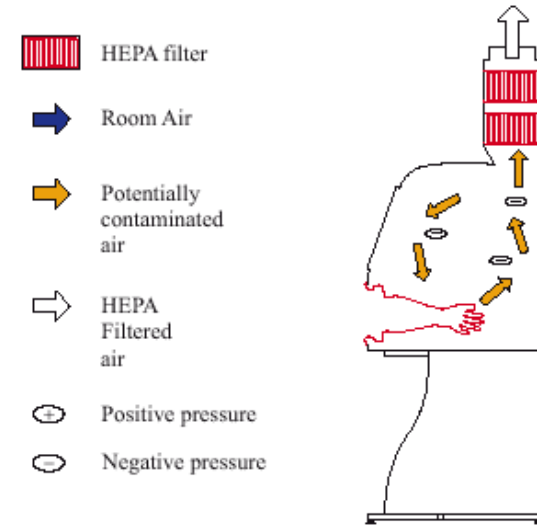
- The bulk fermentation medium
- Nutrient addition feeds, normally comprising one component
- Water that may be added to a previously sterilized fermenter
- Acids and alkalis to control pH
- Buffers—used particularly to control pH
- Antifoams to control foam formation

Filter sterilization of fermenter inlet air

- Aerobic fermentations require the continuous addition of considerable quantities of sterile air
- Air inlet filter systems are designed to remove not only
- Bacteria and fungi, but also viruses—particularly bacteriophages in the case of bacterial fermentations.
- Membrane materials for air filtration must be hydrophobic and thus repel water from the filter surface.

Filtration— in labs

- Pore size 0.22 μm , 0.45 μm
- Pore size 20 nm
- Sterilized or non-sterilized



Filtration– industrial scale

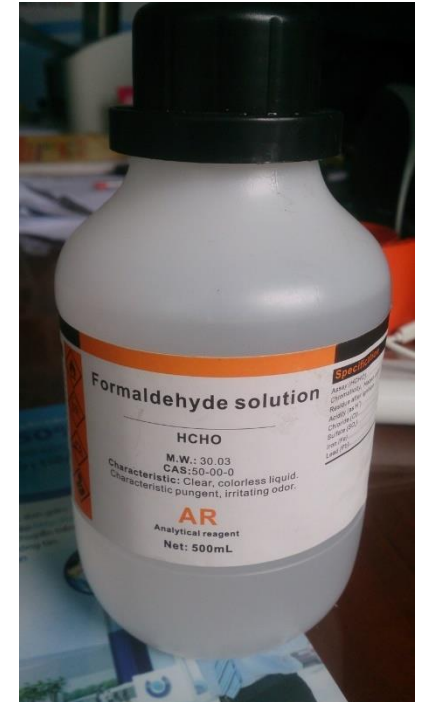
- Air filtration system includes:
 - + Primary filtration: membrane filter with pore sizes 5-30 μm .
 - + Secondary filtration: remove microorganisms, membrane filter with pore sizes 0,1-0,3 μm .

Sterilization of fermenter exhaust air

- In many traditional fermentations, the exhaust gas from the fermenter was vented without sterilization or vented through relatively inefficient depth filters.
- With the advent of the use of recombinant organisms and a greater awareness of safety and emission levels of allergic compounds the containment of exhaust air is virtually universal (and in the case of recombinant organisms, compulsory).

CHEMICAL METHODS

- Use chemicals to eliminate microbial contaminants
- **Advantage:** fast
- **Disadvantage:** limited in applications



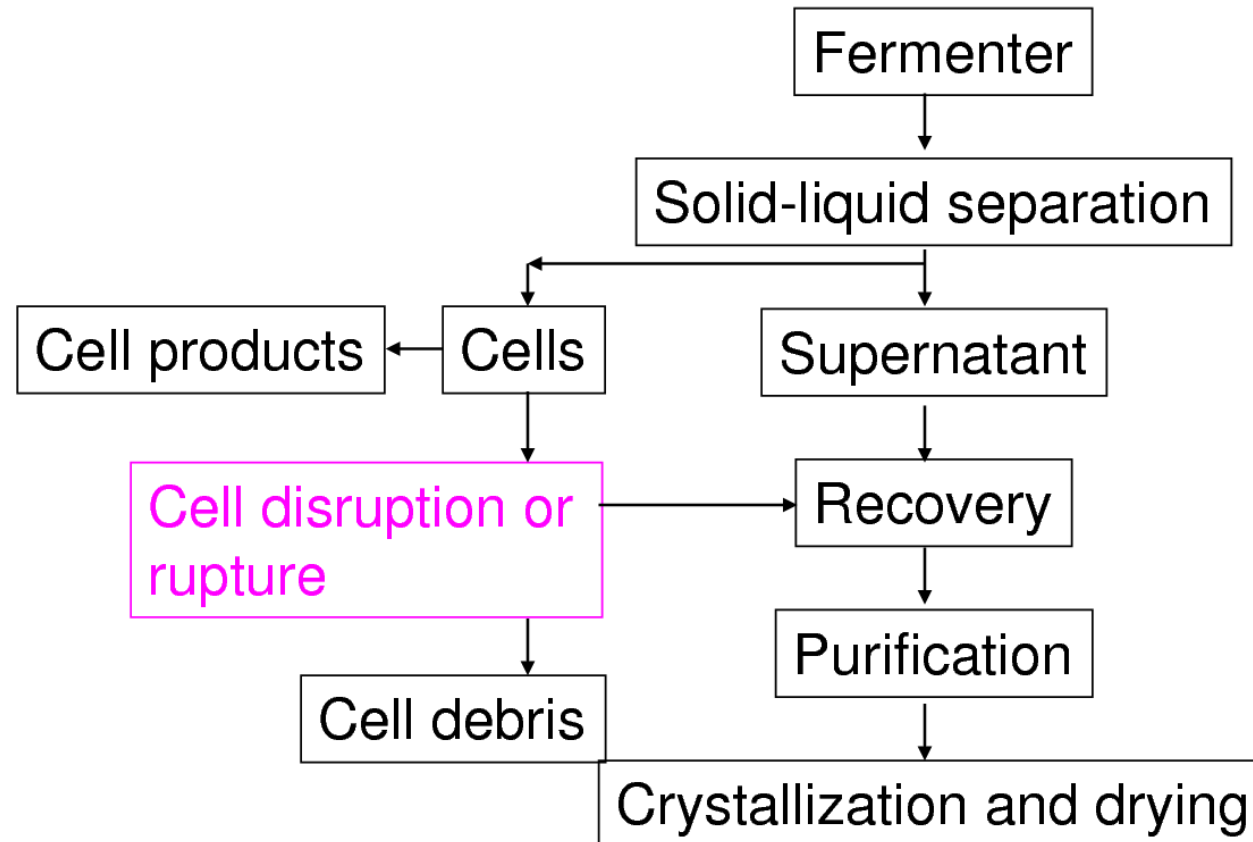
Formaldehyde to sterilize labs

PART 2. RECOVERY OF FERMENTATION PRODUCTS

I. INTRODUCTION

Aims of microbial fermentation

- Microbial cells
- Metabolites: intracellular or extracellular



Costs

- Recovery costs of microbial products may vary from as low as 15% to as high as 70% of the total manufacturing costs

Ex:

Contents in fermentation broth?

- the specific product at a low concentration (typically $0.1\text{--}5\text{ g dm}^{-3}$)
- microorganisms
- cell fragments
- soluble and insoluble medium components
- other metabolic products.

The specific product may also be intracellular, heat labile, and easily broken down by contaminating microorganisms → increase the difficulties of product recovery.

The choice of recovery process is based on the following criteria:

1. Target product: microbial cells or metabolites
2. The intracellular or extracellular location of the product.
3. The concentration of the product in the fermentation broth.
4. The physical and chemical properties of the desired product
5. The intended use of the product.
6. The minimal acceptable standard of purity.
7. The magnitude of biohazard of the product or broth.
8. The impurities in the fermenter broth.
9. The marketable price for the product.

Factors considered to easy the recovery & purification of fermentation products:

1. Selection of a microorganism which does not produce pigments or undesirable metabolites.
2. Modification of the fermentation conditions to reduce the production of undesirable metabolites.
3. Precise timing of harvesting.
4. pH control after harvesting.
5. Temperature treatment after harvesting.
6. Addition of flocculating agents.
7. Use of enzymes to attack cell walls.

- The major problem currently faced in product recovery is the large-scale purification of biologically active molecules.
- The transfer of small-scale preparative/analytical technologies to the production scale while maintaining efficiency of the process, bioactivity of the product and purity of the product.

II. THE RECOVERY OF MICROBIAL CELLS

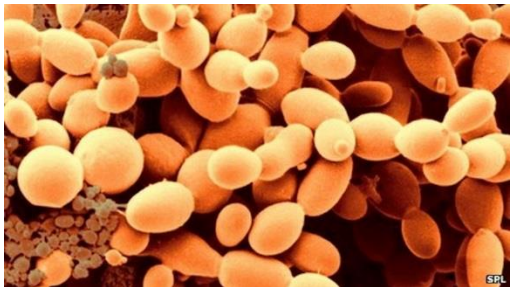
- **Living cells**

Ex: bread yeasts, alcohol yeasts, biological compost, biopesticides,...

- general requirements: living cells and high active state

- **Dead cells (SCP)**

Ex: foods for human and animals,....



- normally separated from the harvested broth by filtration or centrifugation

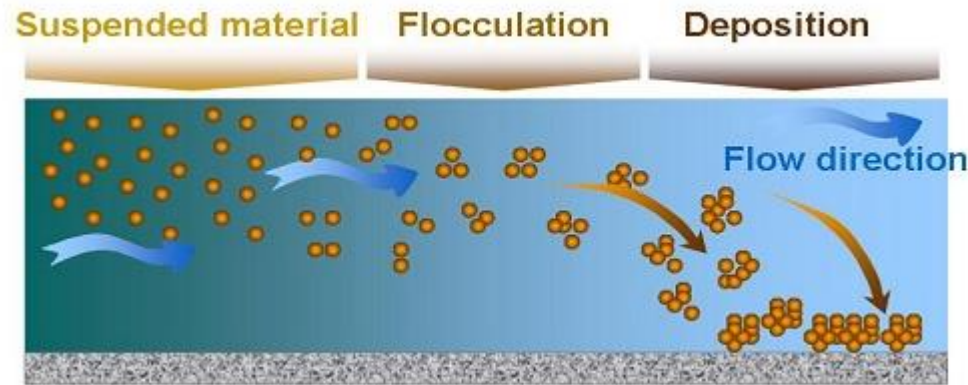
SUPPORTING METHODS/TECHNIQUES

a) Sedimentation (lắng)

- It takes place when particles in suspension settle out of the harvested broth
- **Factors**
 - Conc. of cells, salts, sugars,...
 - Size
 - Viscosity pH, temp., salt of the harvested broth

b) Flocculation (Kết tụ)

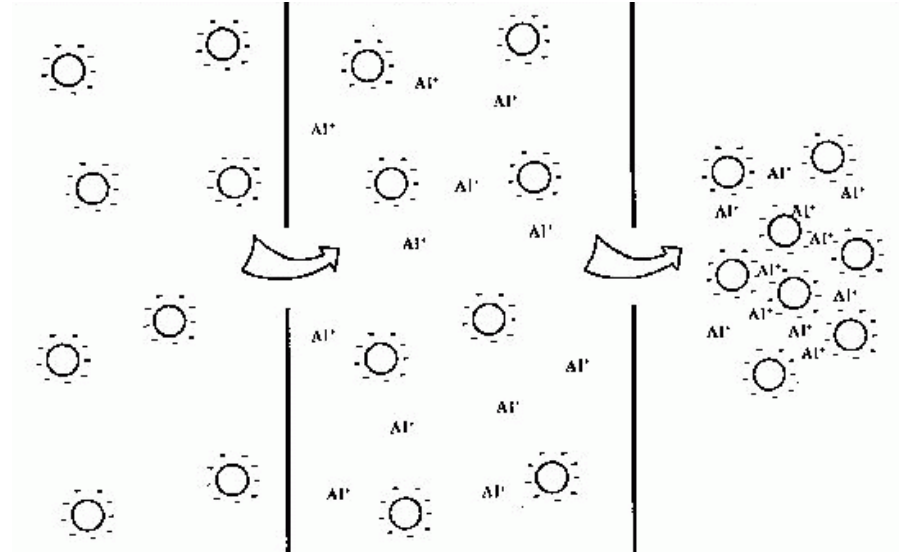
- Flocculation is a process in which particles clump together because the attractive forces between them overcome any repulsive forces caused by like surface charges.
- **Why?**
 - to increase the efficiency of clarification, by making the average particle size larger



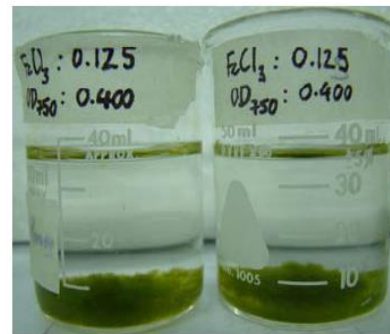
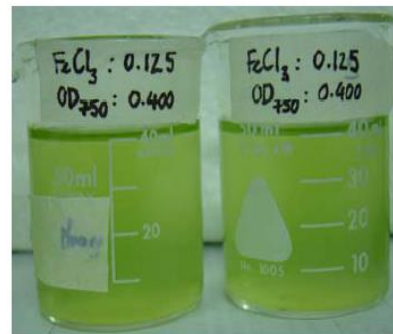
- **Cells can flocculate themselves?**

b) Flocculation (Kết tụ)

- Using positively charged particles

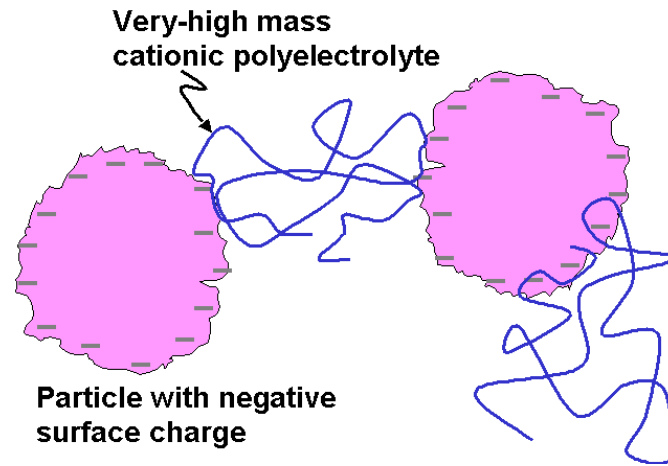


Ex:



b) Flocculation (Kết tụ)

- **Using soluble polymers**
 - Polymers positively charged (+)



- Chitosan...

RECOVERY METHODS

a) Filtration

- One of the most popular methods to recover microbial cells
- Principles: in the previous lecture...



A number of factors:

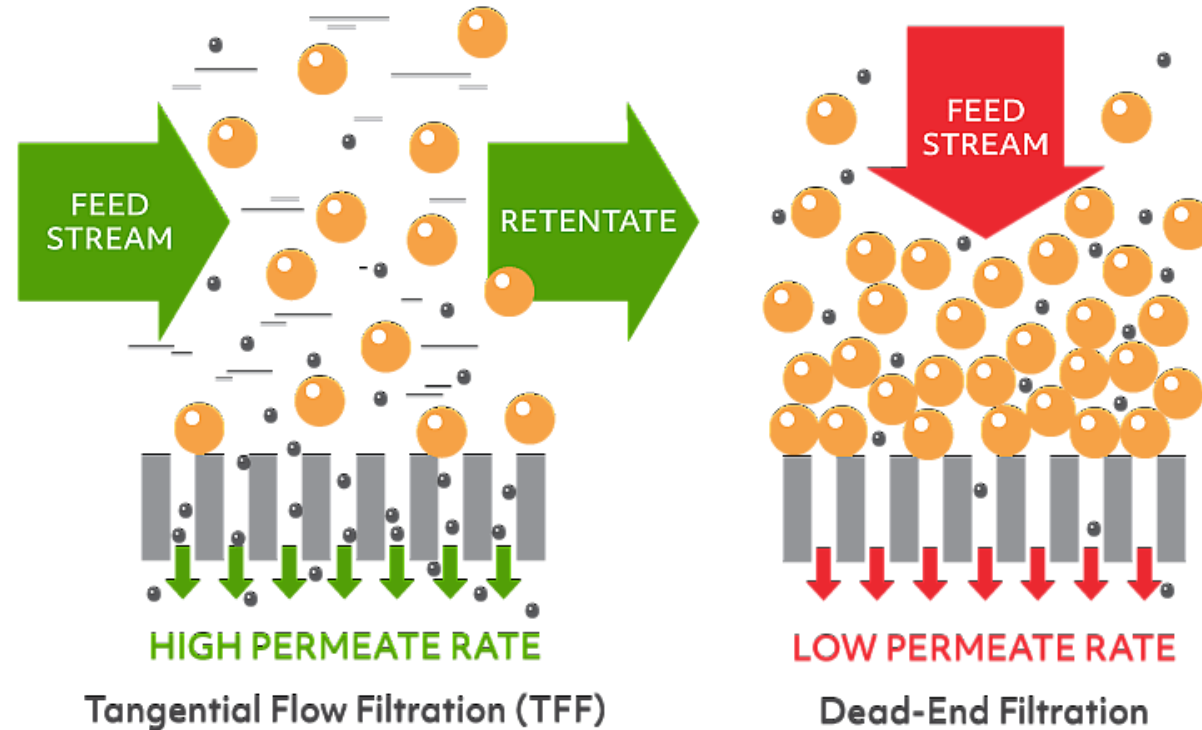
1. The properties of the filtrate, particularly its viscosity and density
2. The nature of the solid particles: size, shape, distribution and packing characteristics.
3. The solids : liquid ratio.
4. The need for recovery of the solid or liquid fraction or both.
5. The scale of operation.
6. The need for batch or continuous operation.
7. The need for aseptic conditions.
8. The need for pressure or vacuum suction to ensure an adequate flow rate of the liquid.

In the filtration processes, if the flow of broth was perpendicular to the filtration membrane
→ blockage of the membrane → lower rates of productivity (serious disadvantages)

RECOVERY METHODS

a) Filtration

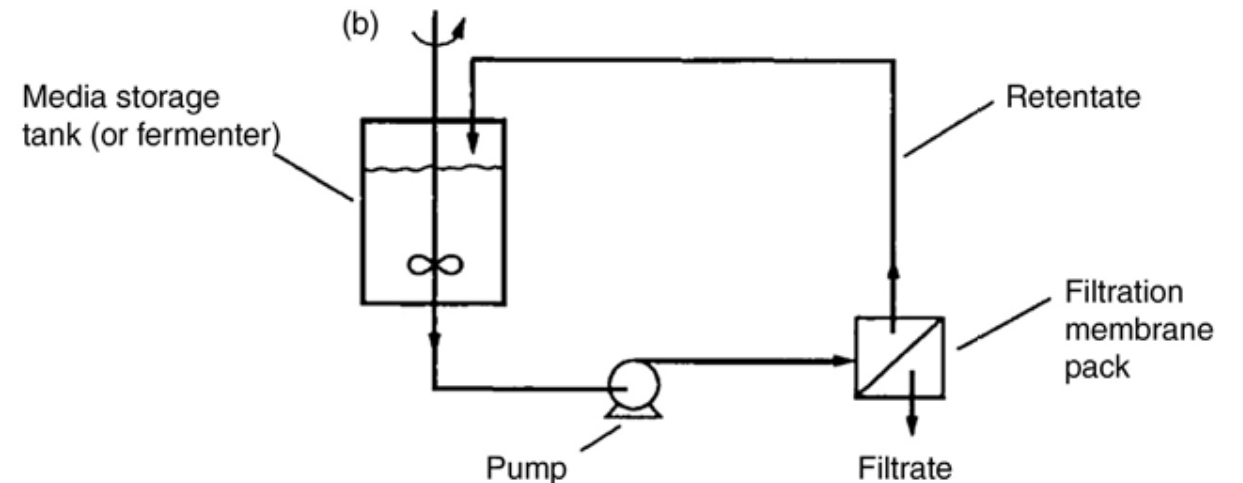
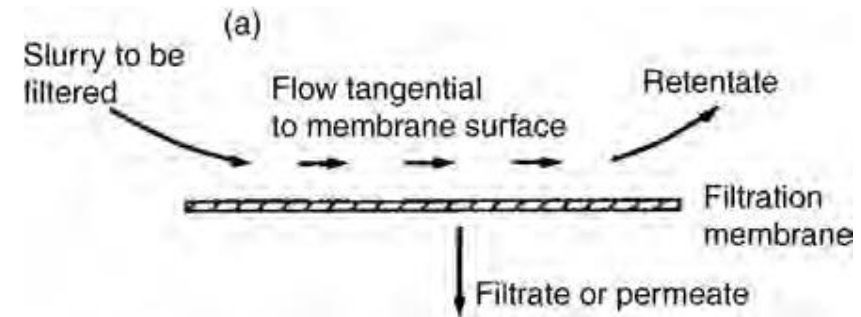
- Cross-flow filtration (tangential filtration)



(image credit: formulatrix.com)

Cross-flow filtration (tangential filtration)

- The major components: a media storage tank (or the fermenter), a pump, and a membrane pack
- Two types of membrane may be used; microporous membranes (microfiltration) with a specific pore size (0.45, 0.22 μm , etc.) or an ultrafiltration membrane (see later section) with a specified molecular weight cut-off (MWCO).



(a) Schematic diagram of cross-flow filtration.
(b) Major components of a cross-flow filtration system.

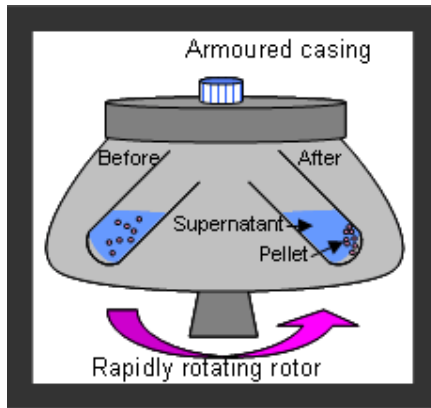
Cross-flow filtration (tangential filtration)

The benefits are:

- 1.** Efficient separation, >99.9% cell retention.
- 2.** Closed system; for the containment of organisms with no aerosol formation
- 3.** Separation is independent of cell and media densities, in contrast to centrifugation.

b) Centrifugation

- More expensive than filtration
- It may be essential when:
 - Filtration is slow and difficult
 - Require high standard of hygiene



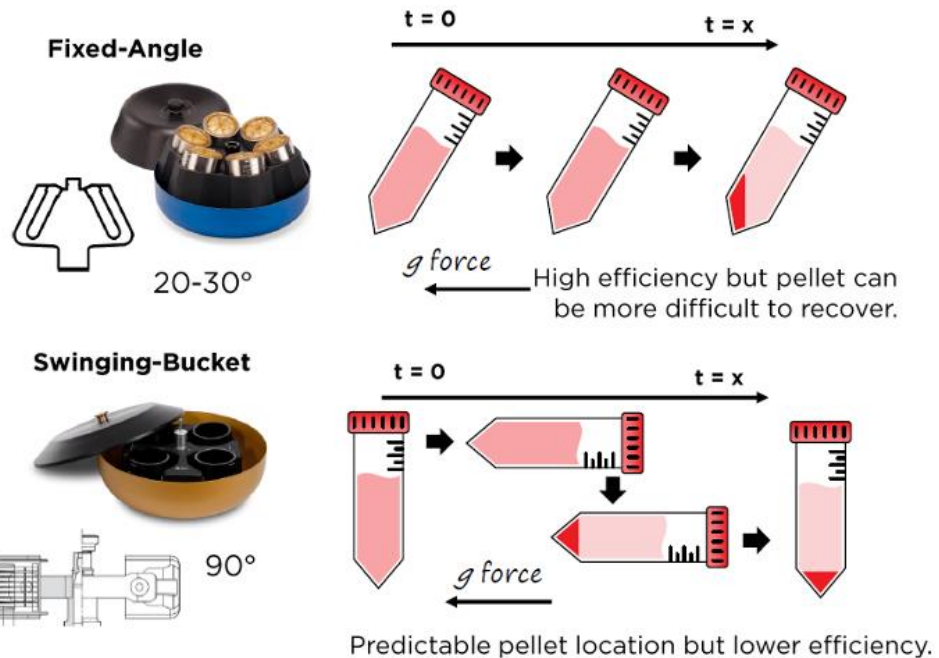
b) Centrifugation

- **RANGE OF CENTRIFUGES**

- *Lab scale*

- + Volume

- + Rotor types

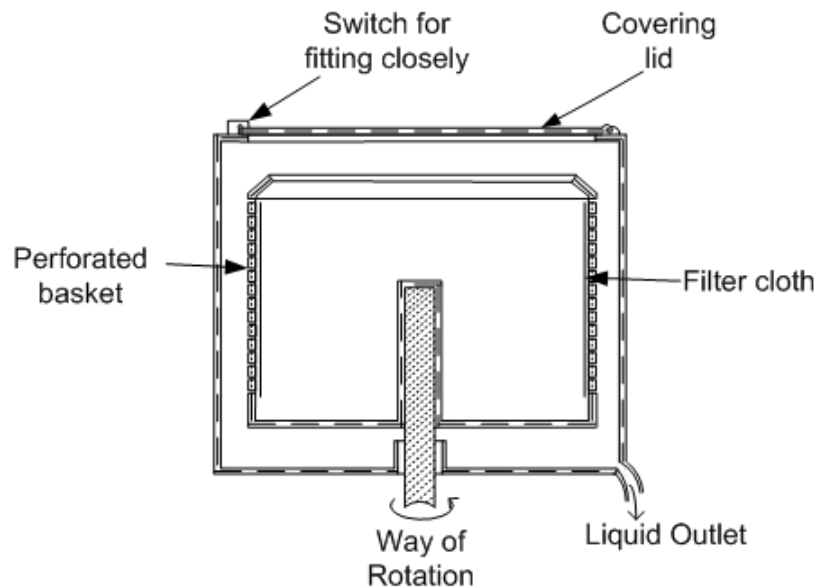


b) Centrifugation

- **RANGE OF CENTRIFUGES**

- *Basket centrifuge*

- + are useful for separating mould mycelia or crystalline compounds



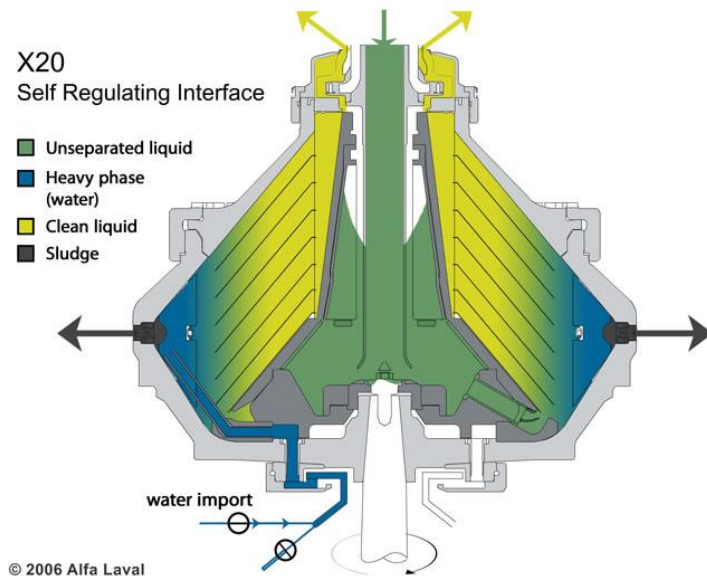
- + Speed: ~ 4000 rpm; Feed rate: $50-300 \text{ dm}^3/\text{min}$

- + $\sim$ Centrifugal filter

b) Centrifugation

- **RANGE OF CENTRIFUGES**

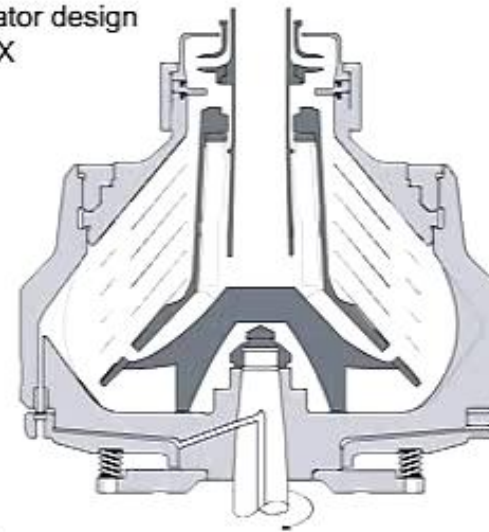
- *Disc bowl centrifuge*



Feed rate: 45-1800 dm³/min

Speed: 5000-10000 rpm

Separator design
type AX

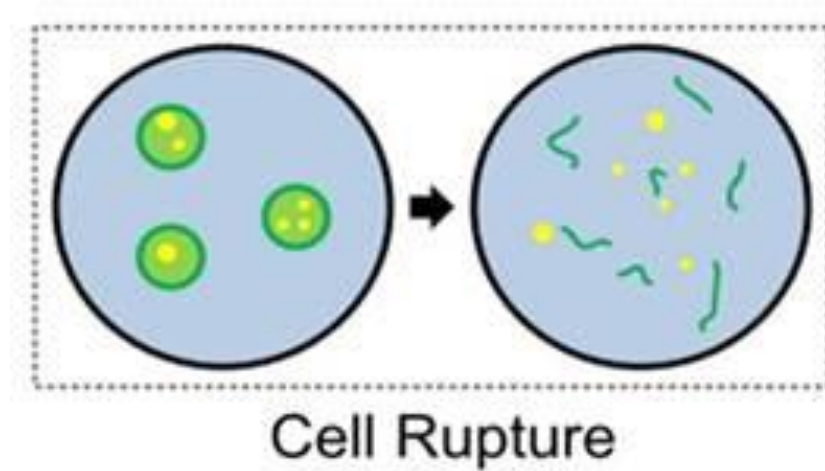


- Type BREW2000; height of 2.2 m; up to 75m³/h



III. RECOVERY OF CELL METABOLITES

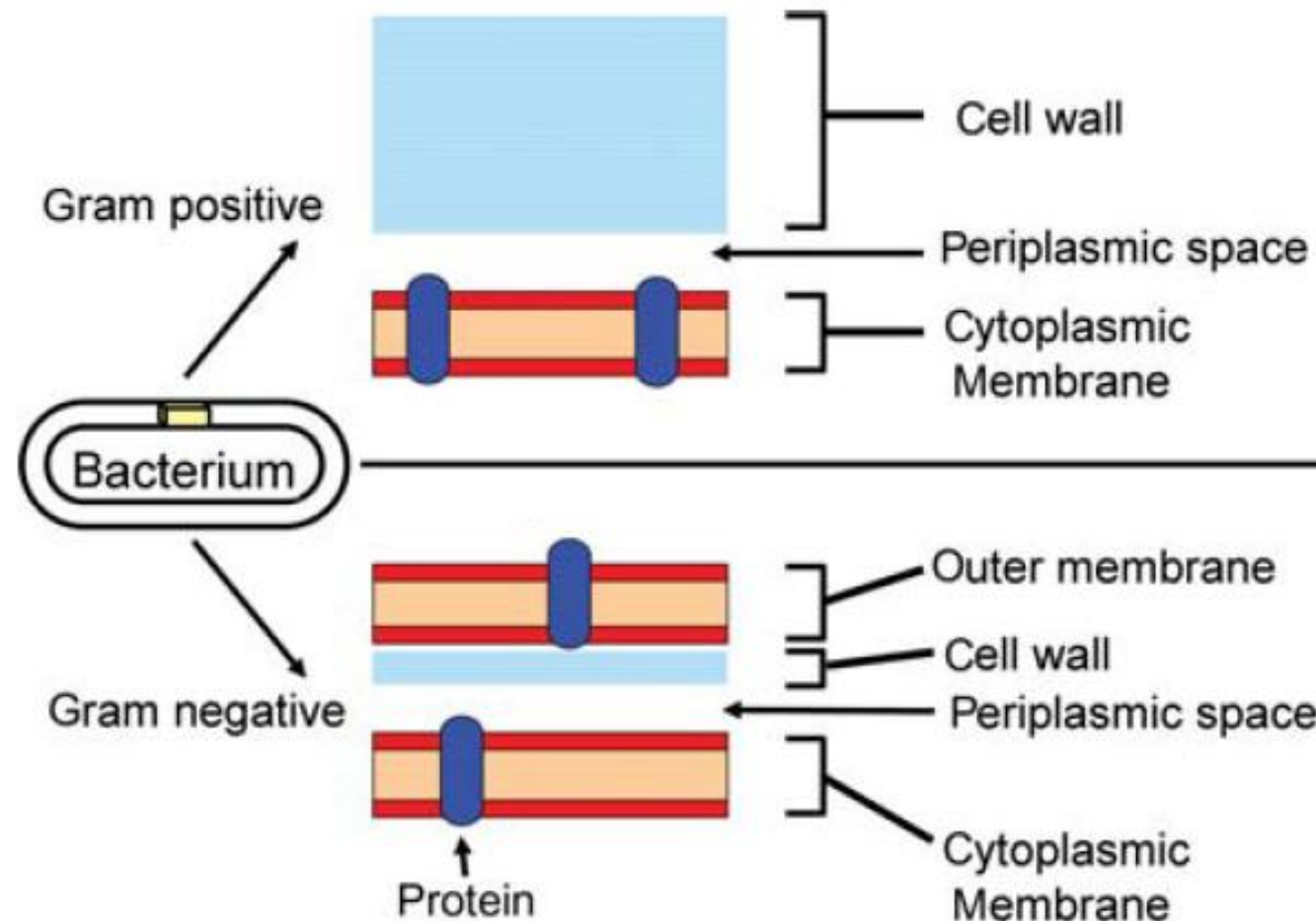
- Gene technology → extracellular target product
- Modify cell wall
- Cell disruption: *mostly used*



CELL DISRUPTION

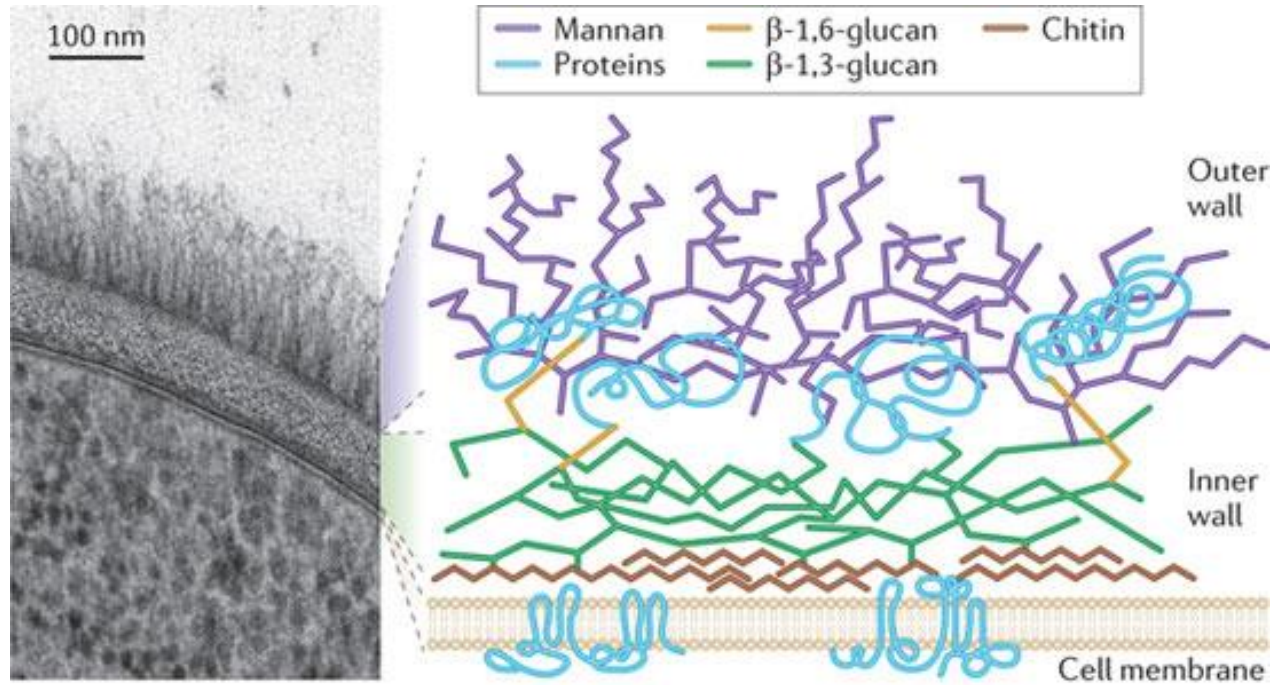
- Microorganisms are protected by extremely tough cell walls.

Cell wall structure of Gram (+) and Gram (-)



Cell wall structure of yeast and fungi

... polysaccharide (100-200 nm)



Micromechanical properties

Organism	Bursting force
<i>Saccharomyces cerevisiae</i>	$101 \pm 24 \mu\text{N}$
<i>Saccharomyces cerevisiae</i>	74–96 μN
<i>Saccharomyces cerevisiae</i>	65–105 μN
<i>Saccharomyces cerevisiae</i>	55–100 μN
<i>Saccharomyces cerevisiae</i>	120–220 μN
<i>Escherichia coli</i>	1–9 μN
<i>Staphylococcus epidermis</i>	3–34 μN

A. niger: ~250–550 μN

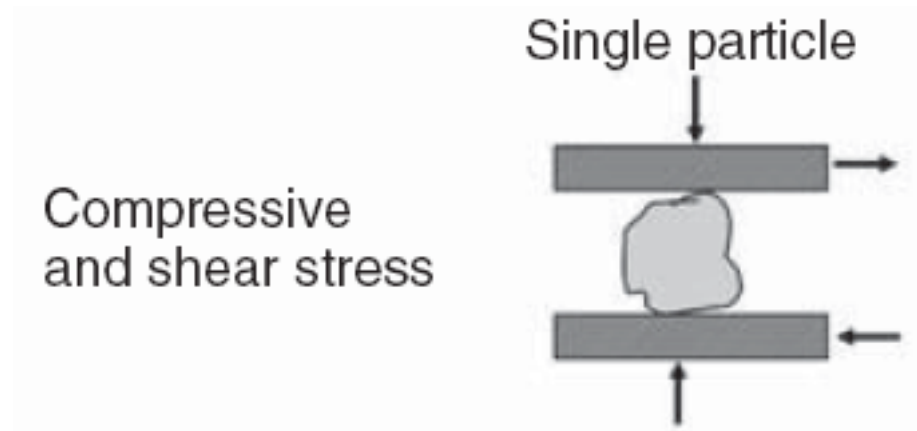
Methods for cell disruption

- Physico-mechanical methods: mechanical stress, high pressure homogenizer, ultrasonication,...
- Chemical methods: Detergents, antibiotics,...
- Biological methods: enzyme, phage,...

Physico-mechanical methods

a) Mechanical stress

- Compressive stress and shear stress

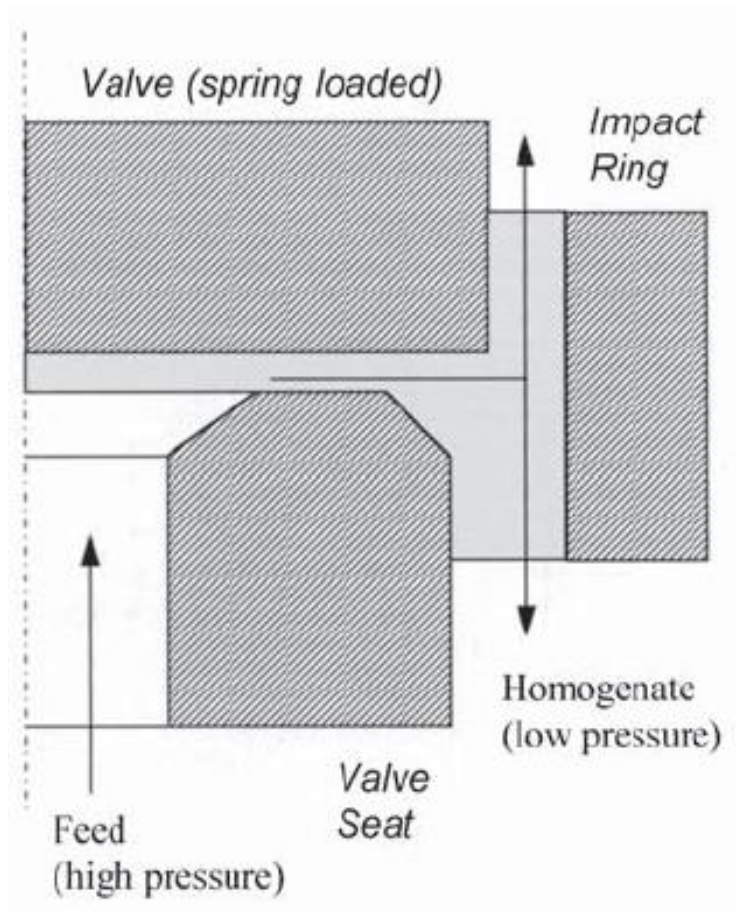


Beads are typically 0.1–3 mm diameter (made of mechanically resistant materials such as glass, alumina ceramics,...)

Advantage: simple

Disadvantage: DNA samples are easily broken...

b) High Pressure Homogenizer (Liquid shear)

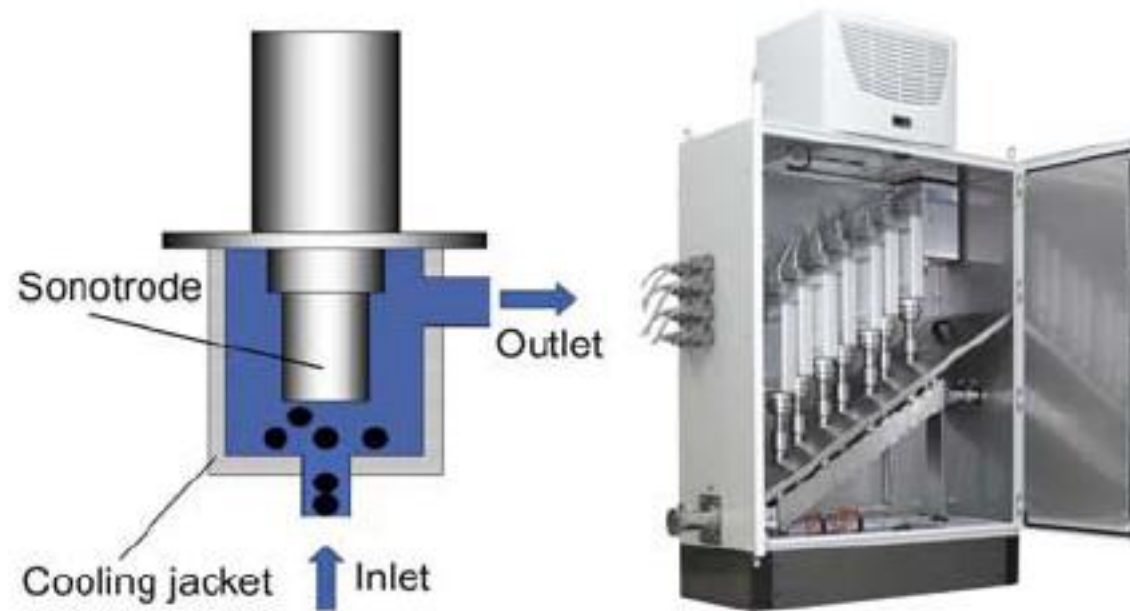


A process-scale homogenizer: GEA Niro Soavi, Ariete NS5180 (3000 L/h at 1500 bar).

- Advantages: high efficiency, high-purified products, large scale
- Disadvantage: expensive

c) Ultrasonication (ultrasonic cavitation)

- High frequency vibration 15–40 kHz → at the tip of an ultrasonication probe leads to cavitation (the formation of vapor cavities in low pressure regions), and shock waves generated when the cavities collapse cause cell disruption

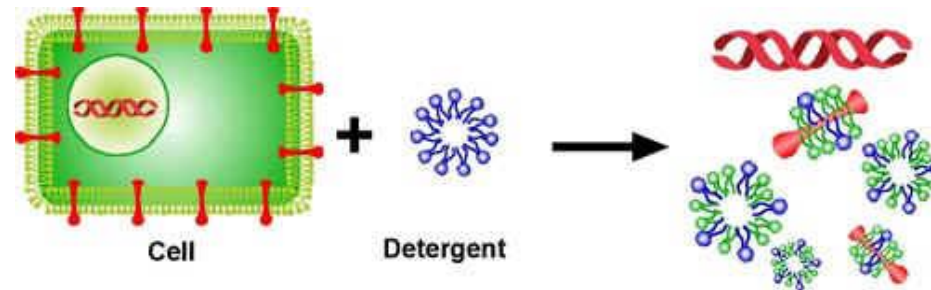


Hielscher Ultrasonics GmbH

- Advantage: high purity
- Disadvantages: high power required, small scale (5–500 mL)

- **Chemical methods**

- Use detergents: NH_4 , sodium lauryl sulfate, sodium dodecyl sulfate (SDS) and Triton X-100,...
- Anionic detergents such as SDS disorganize the cell membrane
- Cationic detergents are believed to act on lipopolysaccharides and phospholipids of the membrane.
- Nonionic detergents such as Triton X-100 cause partial solubilization of membrane proteins



- Advantage: easy to use
- Disadvantages: may cause some protein denaturation
 - : may need to be removed before further purification stages
 - : may contaminate environment

Biological method

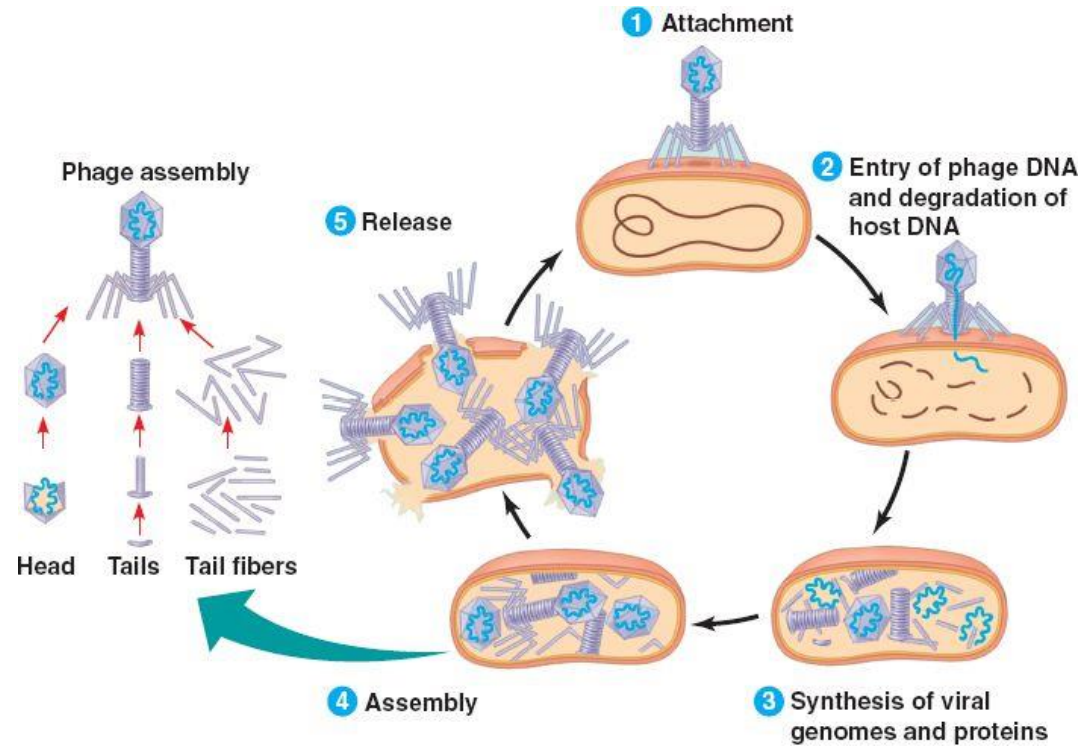
- Enzymes hydrolyze specific bonds in cell wall

Organisms	Enzymes	Target in cell wall
Bacteria	Glycosidases	peptidoglycan
Fungi, yeasts	Glucanases Mannanases Chitinase Proteases	glycans Mannan Chitin Protein
Algae	Cellulases	cellulose

- Advantage: one of the most gentle methods with less affect on product
- Disadvantages: relatively expensive
 - : the presence of the enzyme(s) may complicate further downstream purification processes

Biological method

Use phages



PART 3. PURIFICATION OF FERMENTATION PRODUCTS

- **Cell metabolites normally recovered and purified by:**
 - CHROMATOGRAPHY
 - DISTILLATION
 - MEMBRANE PROCESSES
 - DRYING
 -

CHROMATOGRAPHY

- used to isolate and purify relatively low concentrations of metabolic products
- Chromatographic methods separate solutes based on charge, polarity, size, and affinity
- Chromatography will be concerned with the passage and separation of different solutes as liquid (the mobile phase) is passed through a column, that is, liquid chromatography. Gas chromatography, when the mobile phase is a gas, is a widely used analytical technique but has little application in the recovery of fermentation products.
- Depending on the mechanism by which the solutes may be differentially held in a column, the techniques can be grouped as follows:
 1. Adsorption chromatography.
 2. Ion-exchange chromatography.
 3. Gel permeation chromatography.
 4. Affinity chromatography.
 5. Reverse phase chromatography.
 6. High performance liquid chromatography.

CHROMATOGRAPHY

GEL PERMEATION (gel filtration)

- Separates molecules on the basis of their size
- A wide range of gels are available, including crosslinked dextrans (Sephadex and Sephacryl) and crosslinked agarose (Sepharose) with various pore sizes depending on the fractionation range required
- The smaller molecules diffuse into the gel more rapidly than the larger ones, and penetrate the pores of the gel to a greater degree. This means that once elution is started, the larger molecules which are still in the voids in the gel will be eluted first.

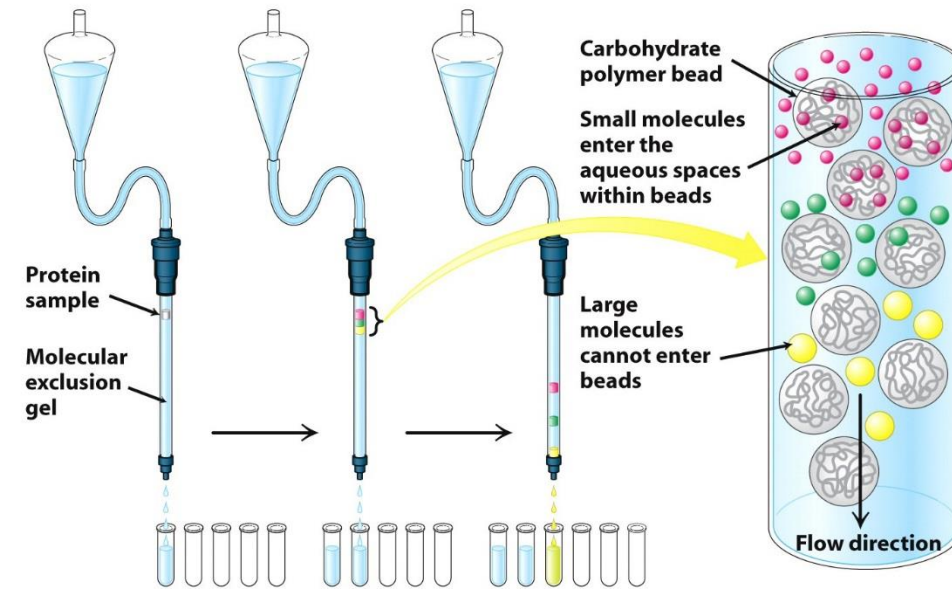


Figure 3.3
Biochemistry, Seventh Edition
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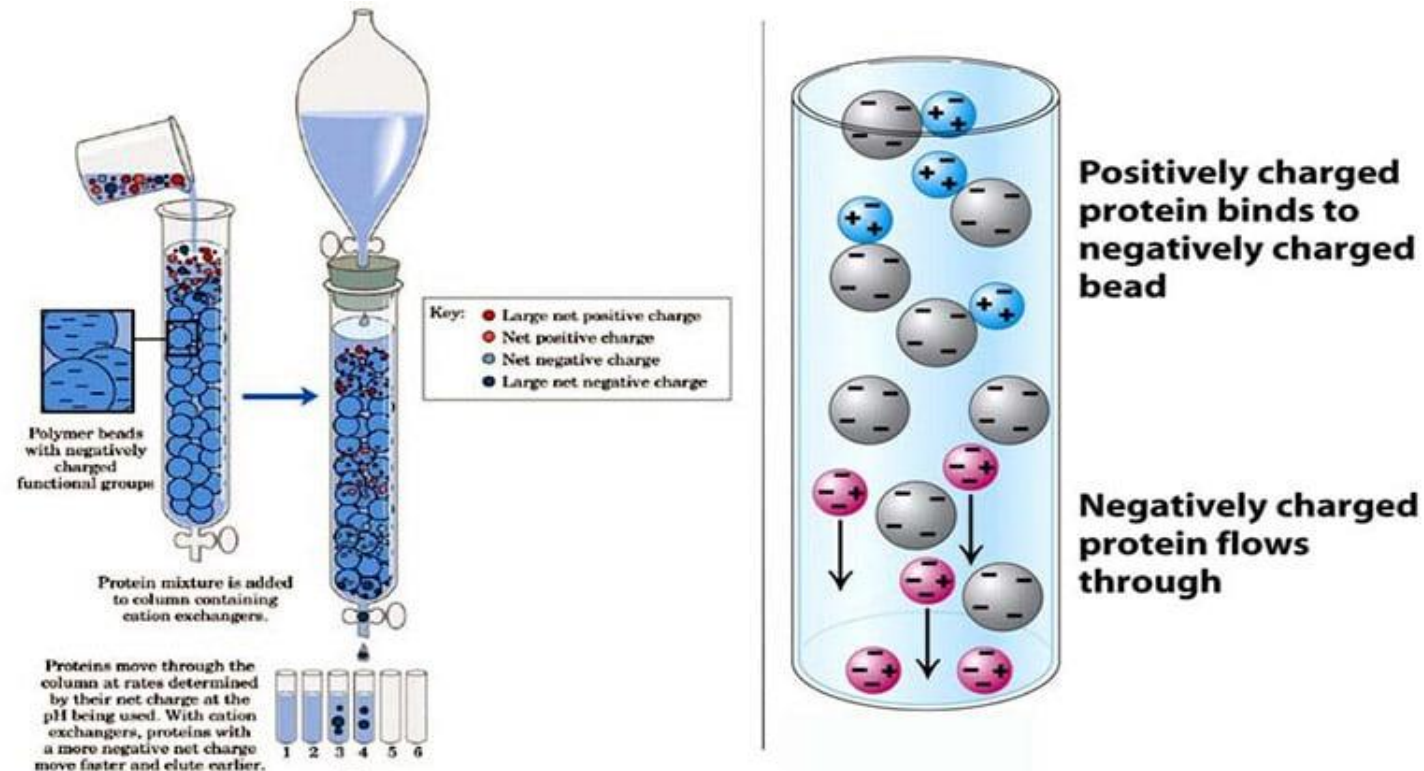
CHROMATOGRAPHY

GEL PERMEATION (gel filtration)

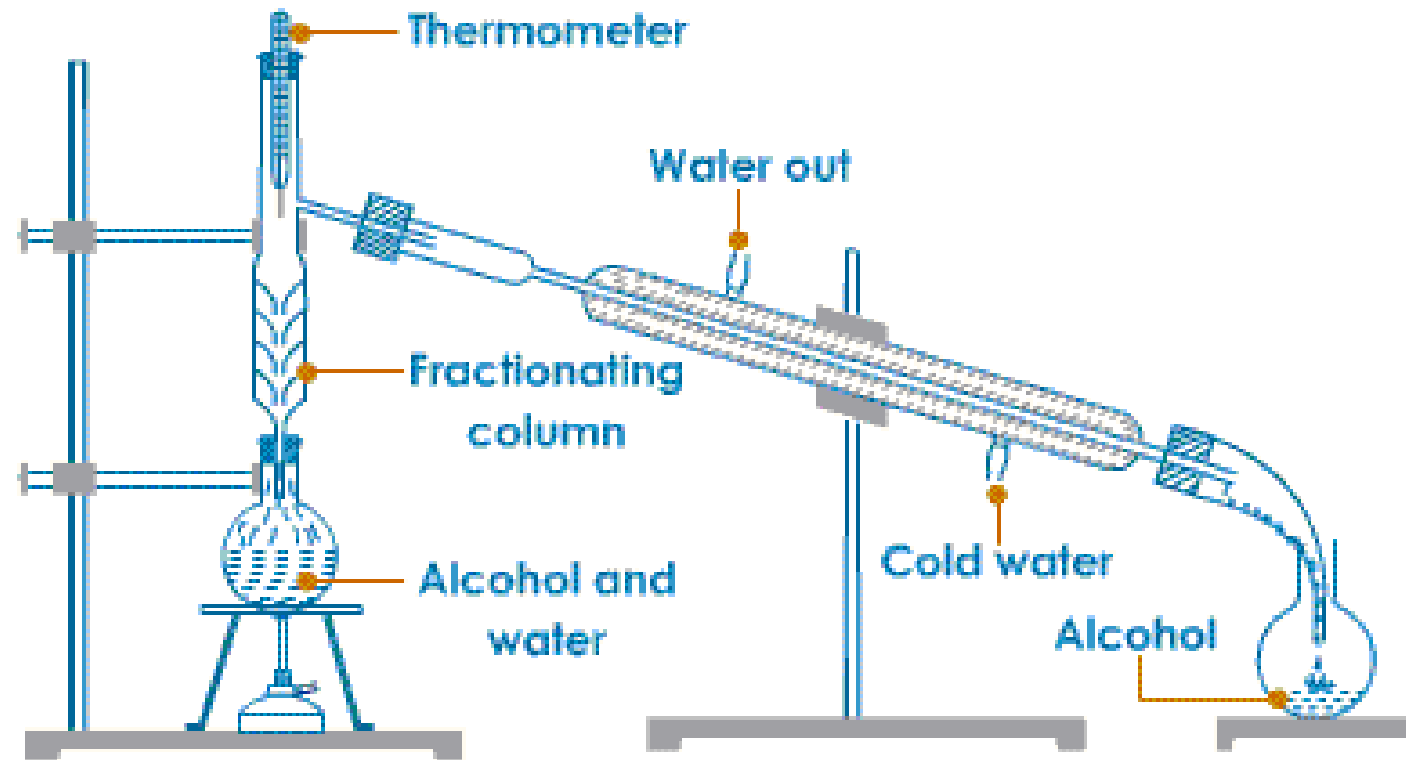
- One early industrial application, although on a relatively small scale, was the purification of vaccines in 1960s
- Possible to be concentrated 10-fold

CHROMATOGRAPHY - ION EXCHANGE

- Reversible exchange of ions between a solid phase (resins normally contain a sulfonic acid, carboxylic acid, or phosphonic acid active group) and a liquid phase (positively charged solutes (eg, certain proteins))
- Positively charged solutes (eg, certain proteins) will bind to the resin, the strength of attachment depending on the net charge of the solute at the pH of the column feed. After deposition solutes are sequentially washed off by the passage of buffers of increasing ionic strength or pH.



Distillation

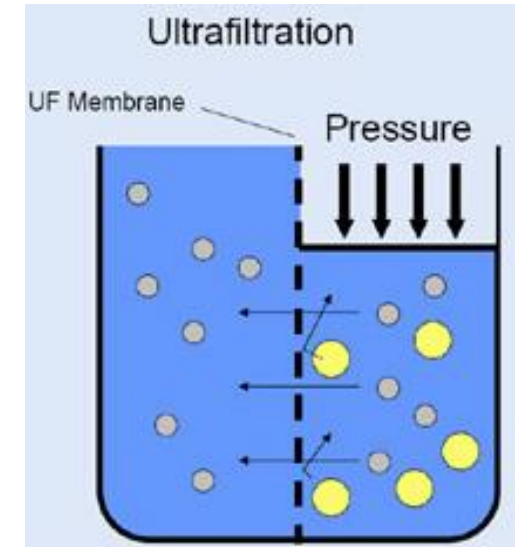


MEMBRANE PROCESSES

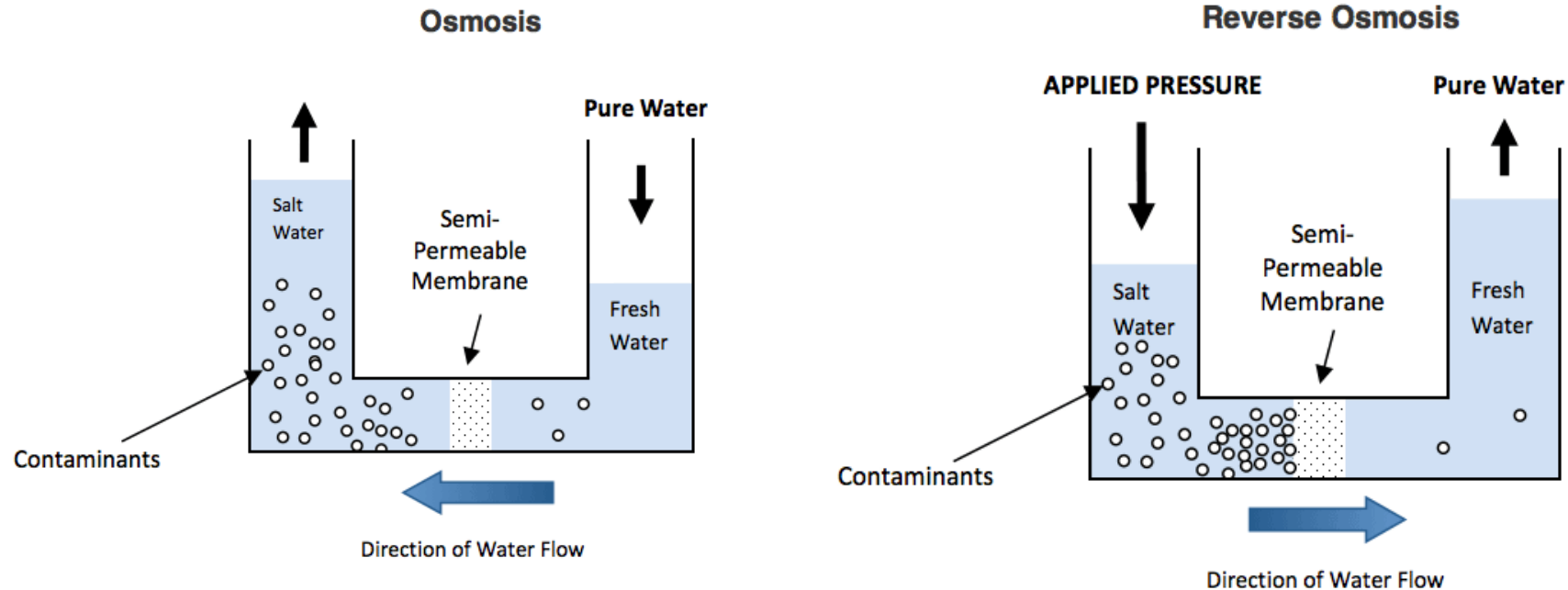
- utilize semipermeable membranes to separate molecules of different sizes
- *2 main methods: ultrafiltration and reverse osmosis*

Ultrafiltration

- solutes of high molecular weight are retained when the solvent and low molecular weight solutes are forced under hydraulic pressure (between 2 and 10 atmospheres) through a membrane of a very fine pore size, typically between 1 and 100 nm → cut-offs (500–500,000 Dalton): separation of macromolecules such as proteins, enzymes, hormones, and viruses.



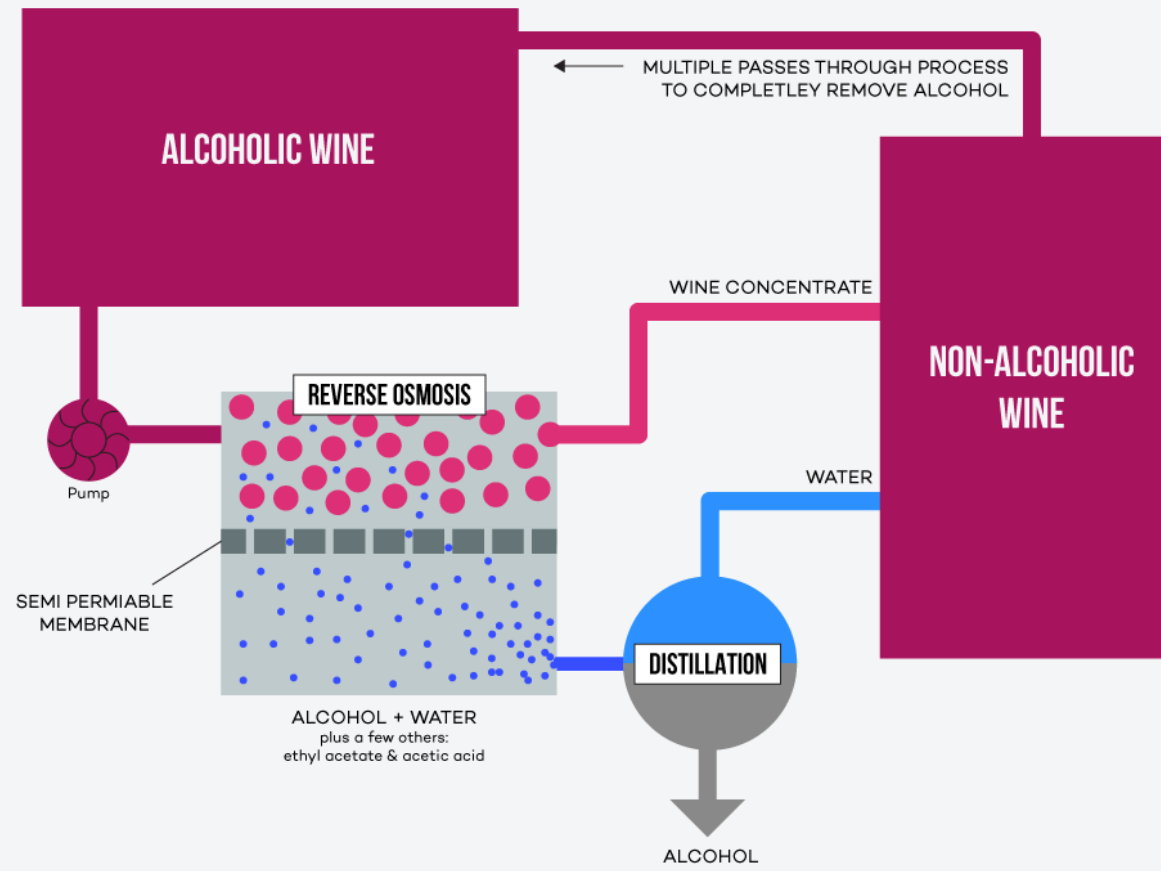
- **Reverse osmosis (hyperfiltration)**



- applied to “small” molecules that are not possible by ultrafiltration

HOW NON-ALCOHOLIC WINE IS MADE

Reverse Osmosis



DRYING

- It involves the final removal of water or other solvents from a product, while ensuring a minimum loss in viability, activity, or nutritional value
- Drying is undertaken because:
 1. The cost of transport can be reduced.
 2. The material is easier to handle and package.
 3. The material can be stored more conveniently in the dry state.

Spray dryer

- is most widely used for drying of biological materials when the starting material is in the form of a liquid or paste
- The material is atomized into small droplets: 10 to 500 μm
- Air temp.:
 - + inlet hot gas 150 – 250 $^{\circ}\text{C}$
 - + exhaust air temp. 75 – 100 $^{\circ}\text{C}$
- Residence times: ~ 0.01 s
- Spray driers are the most economical available for handling large volumes



atomizing nozzle

